

M12x Unit Catalogue

Graphs, mechanisms, tables, and reference hypotheses

Expanded Milestone 1 Part 2 (U1–U7)

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Compiled from mechanism cards (redesign in progress, 2026-07-20)

Abstract

Quick-reference catalogue of the seven accepted M12x units. Each unit is a graph G together with a deterministic mechanism assignment on every non-source. Cells are (unit, $t \in N$, config $\in \{\text{ECAI}, \text{AAMAS}\}$); this document records inputs only (not learner outputs). Units with several targets list each target separately, with its own background-knowledge predictors, labels $E^\pm$, and reference hypothesis $\mathcal{H}_t^*$.

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1 Shared regime

Alphabet

$K = \{0, 1, 2\}$ for every variable.

DGP

Full source factorial over $K^{|S|}$, then propagate mechanisms in topological order. Mechanisms for multi-parent nodes are drawn from $\{\min, \max\}$. Single-parent maps may be unit-specific deterministic functions of the parent (U1: copy; U4: asymmetric indicators). U5 uses the pilot curated (x_0, x_1) support with $x_2 := \text{copy}(x_1)$ (not full source factorial; x_1 is not single-valued in x_0). U7 uses a curated diamond support (noisy fork arms; see §8).

Labels (per target t)

$E^+ = \{t \neq 0\}$, $E^- = \{t = 0\}$ (1-based sample ids on $\mathcal{D}$). Units may later override this per card; U1–U7 keep nonzero-positive.

Background knowledge

Every column except t : exact-value predicates $\text{_val_}0/1/2$ only. No definitional _nz predicates. Descendants of t remain in BK as distractors; citing them in a learned rule is an explicit failure mode.

Reference hypothesis

Compact intensional rules in the `val` language (equationally: the parent conditions that separate $E^\pm$ on $\mathcal{D}$). U5/ $t=x_1$ has no parent-aligned perfect $\mathcal{H}^*$ (descendant only).

Cell count

7 units, 10 (unit, target) pairs, 20 cells after $\times \{\text{ECAI, AAMAS}\}$.

At-a-glance

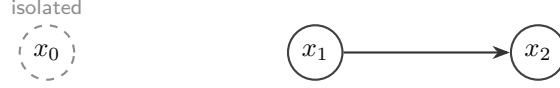
ID	Graph	Mechanisms	Targets N	$ S $	Rows	Cells
U1	separator (+ isolated x_0)	$x_2 := \text{copy}(x_1)$	$\{x_2\}$	2	9	2
U2	collider	$x_2 := \min(x_0, x_1)$	$\{x_2\}$	2	9	2
U3	collider	$x_2 := \max(x_0, x_1)$	$\{x_2\}$	2	9	2
U4	fork	$x_1 := 2 \cdot \mathbf{1}_{x_0 \neq 2}, x_2 := 2 \cdot \mathbf{1}_{x_0 \neq 0}$	$\{x_1, x_2\}$	1	3	4
U5	chain	pilot (x_0, x_1) support; $x_2 := \text{copy}(x_1)$	$\{x_1, x_2\}$	1	6	4
U6	Fabrizio G1	$x_2 := \min; x_3 := x_1 - x_2$	$\{x_2, x_3\}$	2	9	4
U7	diamond	noisy fork arms; $x_3 := x_1 - x_2 $	$\{x_3\}$	1	6	2
10 targets						20

2 U1 — Separator + copy

Fixture key m12_u1_separator_copy
Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$
Learning targets $\{x_2\}$ (single target)

Lesson. Unique parent / copy with an isolated distractor.

2.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_1 \rightarrow x_2\}, \quad S = \{x_0, x_1\}, \quad N = \{x_2\}.$$

2.2 Mechanisms

$$x_2 := \text{copy}(x_1).$$

Isolated x_0 does not enter the mechanism.

2.3 Table $\mathcal{D}$

Full factorial over (x_0, x_1) ; then $x_2 = x_1$.

id	x_0	x_1	$x_2 = \text{copy}(x_1)$
1	0	0	0
2	0	1	1
3	0	2	2
4	1	0	0
5	1	1	1
6	1	2	2
7	2	0	0
8	2	1	1
9	2	2	2

2.4 Target $t = x_2$

Single target — sink

Ancestors $\{x_1\}$; distractor $\{x_0\}$ (isolated, in BK); descendants $\emptyset$.
BK predictors: x_0, x_1 (exclude only x_2 ; **val** only, no ***_nz**).
 $E^+ = \{2, 3, 5, 6, 8, 9\}$ (6 rows, $x_2 \neq 0$); $E^- = \{1, 4, 7\}$ (3 rows, $x_2 = 0$).

Reference $\mathcal{H}_{x_2}^*$:

```
x2(A) :- x1_val_1(A).
x2(A) :- x1_val_2(A).
```

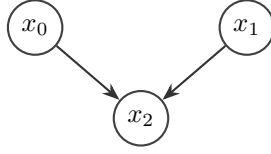
Do not cite isolated x_0 .

3 U2 — Collider + min

Fixture key m12_u2_collider_min
Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$
Learning targets $\{x_2\}$ (single target)

Lesson. Conjunctive / both parents nonzero (min).

3.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_0 \rightarrow x_2, x_1 \rightarrow x_2\}, \quad S = \{x_0, x_1\}, \quad N = \{x_2\}.$$

3.2 Mechanisms

$$x_2 := \min(x_0, x_1).$$

On $\mathcal{D}$: $x_2 \neq 0$ iff both parents are nonzero.

3.3 Table $\mathcal{D}$

id	x_0	x_1	$x_2 = \min$
1	0	0	0
2	0	1	0
3	0	2	0
4	1	0	0
5	1	1	1
6	1	2	1
7	2	0	0
8	2	1	1
9	2	2	2

3.4 Target $t = x_2$

Single target — sink

Ancestors $\{x_0, x_1\}$; other distractors $\emptyset$; descendants $\emptyset$.
 BK predictors: x_0, x_1 (val only, no *_nz).
 $E^+ = \{5, 6, 8, 9\}$ (4 rows); $E^- = \{1, 2, 3, 4, 7\}$ (5 rows).
 On $\mathcal{D}$: $x_2 \neq 0$ iff $x_0 \neq 0$ and $x_1 \neq 0$.

Reference $\mathcal{H}_{x_2}^*$:

```

x2(A) :- x0_val_1(A), x1_val_1(A).
x2(A) :- x0_val_1(A), x1_val_2(A).
x2(A) :- x0_val_2(A), x1_val_1(A).
x2(A) :- x0_val_2(A), x1_val_2(A).

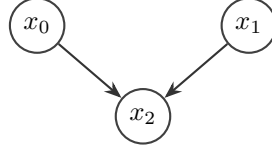
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4 U3 — Collider + max

Fixture key m12_u3_collider_max
Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$
Learning targets $\{x_2\}$ (single target)

Lesson. Disjunctive / either parent nonzero (max).
Same edge set as U2; mechanism differs (max vs min).

4.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_0 \rightarrow x_2, x_1 \rightarrow x_2\}, \quad S = \{x_0, x_1\}, \quad N = \{x_2\}.$$

4.2 Mechanisms

$$x_2 := \max(x_0, x_1).$$

On $\mathcal{D}$: $x_2 \neq 0$ iff at least one parent is nonzero.

4.3 Table $\mathcal{D}$

id	x_0	x_1	$x_2 = \max$
1	0	0	0
2	0	1	1
3	0	2	2
4	1	0	1
5	1	1	1
6	1	2	2
7	2	0	2
8	2	1	2
9	2	2	2

4.4 Target $t = x_2$

Single target — sink

Ancestors $\{x_0, x_1\}$; other distractors $\emptyset$; descendants $\emptyset$.

BK predictors: x_0, x_1 (val only, no *_nz).

$E^+ = \{2, \dots, 9\}$ (8 rows); $E^- = \{1\}$ (1 row).

On $\mathcal{D}$: $x_2 \neq 0$ iff $x_0 \neq 0$ or $x_1 \neq 0$.

Reference $\mathcal{H}_{x_2}^*$:

$x_2(A) \text{ :- } x_0_val_1(A).$

$x_2(A) \text{ :- } x_0_val_2(A).$

$x_2(A) \text{ :- } x_1_val_1(A).$

$x_2(A) \text{ :- } x_1_val_2(A).$

5 U4 — Fork + asymmetric maps

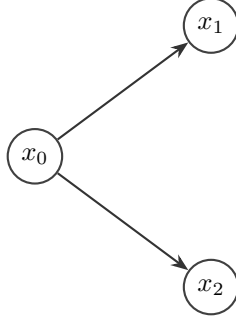
Fixture key $\quad \quad \quad$ m12_u4_fork_double_copy

Sources / rows $\quad |S| = 1 \quad \Rightarrow \quad |\mathcal{D}| = 3$

Learning targets $\quad \{x_1, x_2\}$ — two targets (asymmetric children)

Lesson. Parent vs sibling distractor (sibling correlated but imperfect).

5.1 Graph



$$V = \{x_0, x_1, x_2\}, \quad E = \{x_0 \rightarrow x_1, x_0 \rightarrow x_2\}, \quad S = \{x_0\}, \quad N = \{x_1, x_2\}.$$

5.2 Mechanisms

$$x_1 := \begin{cases} 2 & \text{if } x_0 \neq 2 \\ 0 & \text{otherwise} \end{cases},$$

$$x_2 := \begin{cases} 2 & \text{if } x_0 \neq 0 \\ 0 & \text{otherwise} \end{cases}.$$

Parent uniquely separates each child's nonzero labelling; sibling is imperfect.

5.3 Table $\mathcal{D}$

id	x_0	x_1	x_2
1	0	2	0
2	1	2	2
3	2	0	2

5.4 Target $t = x_1$

Target 1 of 2 — child of x_0

Ancestors $\{x_0\}$; sibling distractor $\{x_2\}$ (**in BK**); descendants $\emptyset$.

BK predictors: x_0, x_2 (**val** only, no ***_nz**).

$E^+ = \{1, 2\}$; $E^- = \{3\}$.

On $\mathcal{D}$: $x_1 \neq 0$ iff $x_0 \in \{0, 1\}$.

Reference $\mathcal{H}_{x_1}^*$:

$x1(A) \text{ :- } x0_val_0(A).$
 $x1(A) \text{ :- } x0_val_1(A).$

Do not cite sibling x_2 .

5.5 Target $t = x_2$

Target 2 of 2 — child of x_0

Ancestors $\{x_0\}$; sibling distractor $\{x_1\}$ (**in BK**); descendants $\emptyset$.

BK predictors: x_0, x_1 (**val** only, no ***_nz**).

$E^+ = \{2, 3\}$; $E^- = \{1\}$.

On $\mathcal{D}$: $x_2 \neq 0$ iff $x_0 \neq 0$.

Reference $\mathcal{H}_{x_2}^*$:

```

x2(A) :- x0_val_1(A).
x2(A) :- x0_val_2(A).

```

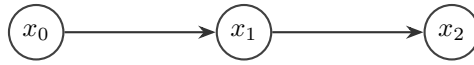
Do not cite sibling x_1 .

6 U5 — Chain + correlated ancestor (pilot)

Fixture key m12_u5_chain_double_copy
Sources / rows $|S| = 1 \Rightarrow |\mathcal{D}| = 6$
Learning targets $\{x_1, x_2\}$ — **two targets** (intermediate + sink)

Lesson. Parent vs correlated ancestor (sink); descendant-in-BK when learning the intermediate.

6.1 Graph



$V = \{x_0, x_1, x_2\}$, $E = \{x_0 \rightarrow x_1, x_1 \rightarrow x_2\}$, $S = \{x_0\}$, $N = \{x_1, x_2\}$.

6.2 Mechanisms

For each $x_0 \in K$, include pairs $(x_0, x_1) = (x_0, x_0)$ and $(x_0, (x_0 + 1) \bmod 3)$; then

$x_2 := \text{copy}(x_1)$.

Not full source factorial: x_1 is not a single-valued function of x_0 . Parent x_1 uniquely separates nonzero labelling of x_2 ; ancestor x_0 is imperfect.

6.3 Table $\mathcal{D}$

id	x_0	x_1	$x_2 = \text{copy}(x_1)$
1	0	0	0
2	0	1	1
3	1	1	1
4	1	2	2
5	2	2	2
6	2	0	0

6.4 Target $t = x_1$ (intermediate)

Target 1 of 2 — intermediate; descendant distractor cell

Ancestors $\{x_0\}$ (imperfect under nonzero labelling); other distractors $\emptyset$; descendants $\{x_2\}$ (**in BK**; citing = failure mode).

BK predictors: x_0, x_2 (val only, no *_nz).

$E^+ = \{2, 3, 4, 5\}$; $E^- = \{1, 6\}$.

On $\mathcal{D}$: $x_1 \neq 0$ iff $x_2 \neq 0$ (because $x_2 = x_1$).

Reference $\mathcal{H}_{x_1}^*$: *None (parent-aligned)*. No perfect $x0_val_*$ separator. Descendant rules are extensionally perfect but are **not** $\mathcal{H}^*$.

6.5 Target $t = x_2$ (sink)

Target 2 of 2 — sink; parent vs ancestor

Ancestors $\{x_0, x_1\}$ (parent x_1 preferred; x_0 correlated but imperfect); descendants $\emptyset$.

BK predictors: x_0, x_1 (val only, no *_nz).

$E^+ = \{2, 3, 4, 5\}$; $E^- = \{1, 6\}$.

On $\mathcal{D}$: $x_2 \neq 0$ iff $x_1 \in \{1, 2\}$.

Reference $\mathcal{H}_{x_2}^*$ (direct parent):

$x_2(A) :- x_1_val_1(A).$

$x_2(A) :- x_1_val_2(A).$

Note: no $x_0_val_*$ -only rule set matches this labelling perfectly. Ancestor citation is a parent-vs-ancestor divergence, not the reference.

7 U6 — Fabrizio G1 + min then difference

Fixture key `m12_u6_g1_and_cone`

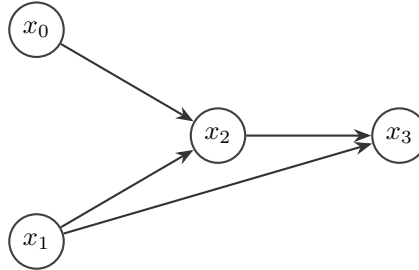
Sources / rows $|S| = 2 \Rightarrow |\mathcal{D}| = 9$

Learning targets $\{x_2, x_3\}$ — two targets (intermediate + sink)

Lesson. Four-node G1; intermediate min (both parents); sink $x_3 \neq 0 \iff x_1 > x_2$ needs both $\text{Pa}(x_3)$; imperfect descendant distractor.

Graph provenance: Fabrizio G1 (ArgCausalDisco four-node example). Mechanisms/DGP are project-chosen.

7.1 Graph



$V = \{x_0, x_1, x_2, x_3\}$, $E = \{x_0 \rightarrow x_2, x_1 \rightarrow x_2, x_1 \rightarrow x_3, x_2 \rightarrow x_3\}$,
 $S = \{x_0, x_1\}$, $N = \{x_2, x_3\}$, $\text{Pa}(x_2) = \{x_0, x_1\}$, $\text{Pa}(x_3) = \{x_1, x_2\}$.

7.2 Mechanisms

$$x_2 := \min(x_0, x_1),$$

$$x_3 := x_1 - x_2.$$

On $\mathcal{D}$, $x_2 \leq x_1$, so $x_3 \in \{0, 1, 2\}$. Sink nonzero labelling: $x_3 \neq 0$ iff $x_1 > x_2$ (both parents required). No $x_3 \equiv x_2$ or $x_3 \equiv x_1$ collapse.

7.3 Table $\mathcal{D}$

id	x_0	x_1	$x_2 = \min(x_0, x_1)$	$x_3 = x_1 - x_2$
1	0	0	0	0
2	0	1	0	1
3	0	2	0	2
4	1	0	0	0
5	1	1	1	0
6	1	2	1	1
7	2	0	0	0
8	2	1	1	0
9	2	2	2	0

7.4 Target $t = x_2$ (intermediate)

Target 1 of 2 — intermediate; descendant x_3 in BK

Ancestors $\{x_0, x_1\}$; descendants $\{x_3\}$ (in **BK**; citing = failure mode).

BK predictors: x_0, x_1, x_3 (val only, no *_nz).

$E^+ = \{5, 6, 8, 9\}$ (4); $E^- = \{1, 2, 3, 4, 7\}$ (5).

On $\mathcal{D}$: $x_2 \neq 0$ iff $x_0 \neq 0$ and $x_1 \neq 0$. Descendant $x_3 \neq 0$ only on $\{2, 3, 6\}$ (imperfect separator).

Reference $\mathcal{H}_{x_2}^*$:

$x2(A) :- x0_val_1(A), x1_val_1(A).$

$x2(A) :- x0_val_1(A), x1_val_2(A).$

$x2(A) :- x0_val_2(A), x1_val_1(A).$

$x2(A) :- x0_val_2(A), x1_val_2(A).$

Must not cite descendant x_3 .

7.5 Target $t = x_3$ (sink)

Target 2 of 2 — sink; $x_1 > x_2$

Ancestors $\{x_0, x_1, x_2\}$; parents $\{x_1, x_2\}$; descendants $\emptyset$.

BK predictors: x_0, x_1, x_2 (val only, no *_nz).

$E^+ = \{2, 3, 6\}$ (3); $E^- = \{1, 4, 5, 7, 8, 9\}$ (6).

On $\mathcal{D}$: $x_3 \neq 0$ iff $x_1 > x_2$ (neither parent alone separates).

Reference $\mathcal{H}_{x_3}^*$ (parent-based):

$x3(A) :- x1_val_1(A), x2_val_0(A).$

$x3(A) :- x1_val_2(A), x2_val_0(A).$

$x3(A) :- x1_val_2(A), x2_val_1(A).$

8 U7 — Diamond + noisy fork arms

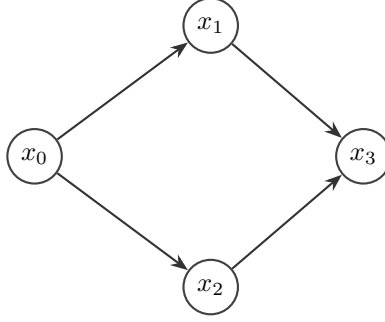
Fixture key m12_u7_g1_or_cone

Sources / rows $|S| = 1 \Rightarrow |\mathcal{D}| = 6$

Learning targets $\{x_3\}$ (single target — sink)

Lesson. Both-parent sink when parents are correlated siblings; root and each sibling alone imperfect (U5-style curated noisy arms).

8.1 Graph



$V = \{x_0, x_1, x_2, x_3\}$, $E = \{x_0 \rightarrow x_1, x_0 \rightarrow x_2, x_1 \rightarrow x_3, x_2 \rightarrow x_3\}$,
 $S = \{x_0\}$, $N = \{x_1, x_2, x_3\}$, $\text{Pa}(x_3) = \{x_1, x_2\}$.

8.2 Mechanisms

For each $x_0 \in K$: $x_1 := (x_0 + 1) \bmod 3$; two x_2 values $(x_0 + 2) \bmod 3$ (differ) and x_1 (collapse); then

$$x_3 := |x_1 - x_2|.$$

Not full source factorial. On $\mathcal{D}$: $x_3 \neq 0$ iff $x_1 \neq x_2$. No singleton among $\{x_0, x_1, x_2\}$ is a perfect val-union separator of $E^\pm$.

8.3 Table $\mathcal{D}$

id	x_0	x_1	x_2	$x_3 = x_1 - x_2 $
1	0	1	2	1
2	0	1	1	0
3	1	2	0	2
4	1	2	2	0
5	2	0	1	1
6	2	0	0	0

8.4 Target $t = x_3$ (sink)

Single target — both sibling parents

Ancestors $\{x_0, x_1, x_2\}$; parents $\{x_1, x_2\}$; descendants $\emptyset$.

BK predictors: x_0, x_1, x_2 (val only, no *_nz).

$E^+ = \{1, 3, 5\}$; $E^- = \{2, 4, 6\}$.

On $\mathcal{D}$: $x_3 \neq 0$ iff $x_1 \neq x_2$.

Reference $\mathcal{H}_{x_3}^*$:

```

x3(A) :- x1_val_1(A), x2_val_2(A).
x3(A) :- x1_val_2(A), x2_val_0(A).
x3(A) :- x1_val_0(A), x2_val_1(A).

```

Source pointers

Mechanism cards

docs/research/milestone_plans/milestone1_part2/mechanism_cards/

Unit set

milestone1_part2_expanded_unit_set.md

Approach

milestone1_part2_expanded_approach.md

Implementation

causal/experiments/handcrafted_m12x.py